

A hyperbolic approximation of the Cahn-Hilliard equation

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UNIVERSITÀ
DI TRENTO



Finanziato
dall'Unione europea
NextGenerationEU

Dec 15th, 2025

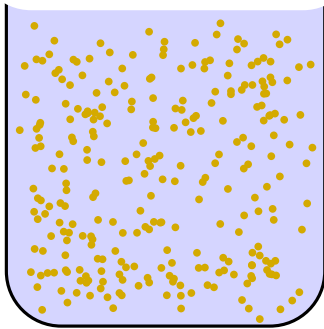
Cahn-Hilliard equations (1958)

The Cahn-Hilliard equation is postulated as a conservative diffusion equation which writes

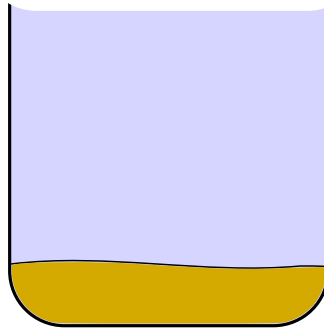
$$\frac{\partial c}{\partial t} = \Delta (c^3 - c - \gamma \Delta c) .$$

- $c \in [-1, 1]$ is the order parameter indicating the phases.
- $\gamma \ll 1$ is such that $\sqrt{\gamma}$ is the diffuse interface characteristic length.
- describes well the process of phase separation in binary systems: spinodal decomposition, Ostwald Ripening phenomena, etc
- Has applications for modeling binary alloys, sedimentation problems, etc ...

Example applications: sedimentation



Dispersed phase



Separated phases

About the equation

$$\frac{\partial c}{\partial t} = \Delta (c^3 - c - \gamma \Delta c) .$$

Cool features

- scalar PDE.
- well-posed [Elliott& Songmu 86¹].
- diffuse-interface model (able to deal with strong topological changes).

¹Elliott, Charles M., and Zheng Songmu. "On the cahn-hilliard equation." Archive for Rational Mechanics and Analysis 96.4 (1986): 339-357.

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- scalar PDE.
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- diffuse-interface model (able to deal with strong topological changes).

Not so cool features

- non-convex equation of state (Requires very careful numerical treatment)
- 4th Order in space (Forget about explicit solvers)
- Violates principle of Causality (Laplace operator)

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Plan of presentation

- 1 On the Cahn-Hilliard equations
- 2 Hyperbolic Model Derivation
 - 2nd-order approximation
 - 1st-order approximation approximation
 - Analysis
- 3 Numerical scheme and Results
 - Numerical scheme
 - Some results

Conservative form and chemical potential

The Cahn-Hilliard equation can be cast into a conservation-law form which writes

$$\frac{\partial c}{\partial t} - \Delta (c^3 - c - \gamma \Delta c) = 0 \quad \Rightarrow \quad \frac{\partial c}{\partial t} + \operatorname{div}(\mathbf{j}) = 0,$$

where the mass flux $\mathbf{j}$ is assumed to obey a generalized Fick's law such that

$$\mathbf{j} = -\nabla \mu[c],$$

and μ is the chemical potential of the system given by

$$\mu[c] = \frac{\delta f}{\delta c} = \frac{\partial f}{\partial c} - \operatorname{div} \left(\frac{\partial f}{\partial \nabla c} \right) = c^3 - c - \gamma \Delta c,$$

where

$$f(c, \nabla c) = \frac{(c^2 - 1)^2}{4} + \frac{\gamma}{2} \|\nabla c\|^2,$$

Lyapunov functional

CH equation admits the Lyapunov functional

$$F(c, \nabla c) = \int_{\mathcal{D}} f(c, \nabla c) \, d\Omega. \quad f(c, \nabla c) = \frac{(c^2 - 1)^2}{4} + \frac{\gamma}{2} \|\nabla c\|^2,$$

Indeed, we have

$$\frac{\partial f}{\partial t} + \operatorname{div}(\mathcal{H}) = -\|\nabla \mu[c]\|^2, \quad \mathcal{H} = -\mu[c] \nabla \mu[c] - \frac{\partial f}{\partial \nabla c} \Delta \mu[c],$$

which in integral form writes

$$\frac{\partial F}{\partial t} = - \int_{\mathcal{D}} \|\nabla \mu[c]\|^2 \, d\Omega \leq 0.$$

Hyperbolic reformulation

- 1 On the Cahn-Hilliard equations
- 2 **Hyperbolic Model Derivation**
 - 2nd-order approximation
 - 1st-order approximation approximation
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Modified action functional

Let us introduce the following action functional

$$a = \int_t \int_{\mathcal{D}} \mathcal{L} \, d\mathcal{D} dt$$

where

$$\mathcal{L} \left(c, \varphi, \nabla \varphi, \frac{\partial \varphi}{\partial t} \right) = -\frac{(c^2 - 1)^2}{4} - \frac{\gamma}{2} \|\nabla \varphi\|^2 - \frac{\alpha}{2} (c - \varphi)^2 + \frac{\beta}{2} \left(\frac{\partial \varphi}{\partial t} \right)^2.$$

- φ is a new variable substituting c as the order parameter.
- $\alpha \gg 1$ so that $(c - \varphi)$ vanishes in the limit $\alpha \rightarrow +\infty$.
- $\beta \ll 1$ is a small parameter.

Generalized Fick's law for c

$$\mathcal{L}\left(c, \varphi, \nabla \varphi, \frac{\partial \varphi}{\partial t}\right) = -\frac{(c^2 - 1)^2}{4} - \frac{\gamma}{2} \|\nabla \varphi\|^2 - \frac{\alpha}{2}(c - \varphi)^2 + \frac{\beta}{2} \left(\frac{\partial \varphi}{\partial t}\right)^2.$$

Generalized Fick's law now becomes

$$\frac{\partial c}{\partial t} + \operatorname{div}(-\nabla \mu) = 0, \quad \mu = -\frac{\delta \mathcal{L}}{\delta c} = -\frac{\partial \mathcal{L}}{\partial c} = c^3 - c + \alpha(c - \varphi),$$

$\Rightarrow$ 2nd-order PDE, no 4th-order terms

$$\frac{\partial c}{\partial t} - \Delta (c^3 - c + \alpha(c - \varphi)) = 0, \quad (I)$$

Euler-Lagrange equation for φ

$$\mathcal{L}\left(c, \varphi, \nabla \varphi, \frac{\partial \varphi}{\partial t}\right) = -\frac{(c^2 - 1)^2}{4} - \frac{\gamma}{2} \|\nabla \varphi\|^2 - \frac{\alpha}{2}(c - \varphi)^2 + \frac{\beta}{2} \left(\frac{\partial \varphi}{\partial t}\right)^2.$$

For φ , we simply write the Euler-Lagrange equations.

$$\frac{\partial}{\partial t} \left(\frac{\partial \mathcal{L}}{\partial \varphi_t} \right) + \operatorname{div} \left(\frac{\partial \mathcal{L}}{\partial \nabla \varphi} \right) = \frac{\partial \mathcal{L}}{\partial \varphi}.$$

which gives

$$\beta \frac{\partial^2 \varphi}{\partial t^2} - \operatorname{div}(\gamma \nabla \varphi) = \alpha(c - \varphi) \quad (II)$$

2nd-order approximation of the Cahn-Hilliard equation

Thus so far we have obtained the following system of two 2nd order PDEs

$$\begin{aligned}\frac{\partial c}{\partial t} - \operatorname{div} (\nabla (c^3 - c + \alpha(c - \varphi))) &= 0, \\ \beta \frac{\partial^2 \varphi}{\partial t^2} - \operatorname{div} (\gamma \nabla \varphi) &= \alpha(c - \varphi).\end{aligned}$$

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- Equation (I) is reminiscent of heat equation.
 $\Rightarrow$ Cattaneo-type relaxation.

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- Equation (I) is reminiscent of heat equation.
 $\Rightarrow$ Cattaneo-type relaxation.
- Equation (II) is a hyperbolic wave equation with right-hand side. $\Rightarrow$ Order reduction.

Order reduction for (II)

$$\beta \frac{\partial^2 \varphi}{\partial t^2} - \operatorname{div}(\gamma \nabla \varphi) = \alpha(c - \varphi) \quad (II)$$

Let us denote the independent variables

$$w = \beta \frac{\partial \varphi}{\partial t}, \quad \mathbf{p} = \nabla \varphi.$$

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Let us denote the independent variables

$$w = \beta \frac{\partial \varphi}{\partial t}, \quad \mathbf{p} = \nabla \varphi.$$

Therefore (II) becomes

$$\begin{aligned} \frac{\partial w}{\partial t} - \operatorname{div}(\gamma \mathbf{p}) &= -\alpha(\varphi - c), \\ \frac{\partial \varphi}{\partial t} &= \frac{1}{\beta} w, \\ \frac{\partial \mathbf{p}}{\partial t} - \frac{1}{\beta} \nabla w &= 0. \end{aligned}$$

Relaxation for equation (I)

$$\begin{aligned}\frac{\partial c}{\partial t} + \operatorname{div} \left(\frac{1}{\tau} \mathbf{q} \right) &= 0, \\ \frac{\partial \mathbf{q}}{\partial t} + \nabla \mu &= -\frac{1}{\tau} \mathbf{q},\end{aligned}$$

- $\tau \ll 1$ is a relaxation time.
- c is still a conserved quantity in this framework.

Final system approximating the Cahn-Hilliard equations

$$\frac{\partial c}{\partial t} + \operatorname{div} \left(\frac{1}{\tau} \mathbf{q} \right) = 0$$

$$\frac{\partial \mathbf{q}}{\partial t} + \nabla (c^3 - c + \alpha(c - \varphi)) = -\frac{1}{\tau} \mathbf{q}$$

$$\frac{\partial w}{\partial t} - \operatorname{div} (\gamma \mathbf{p}) = -\alpha(\varphi - c)$$

$$\frac{\partial \mathbf{p}}{\partial t} - \frac{1}{\beta} \nabla w = 0$$

$$\frac{\partial \varphi}{\partial t} = \frac{1}{\beta} w$$

- System of hyperbolic equations with relaxations.
- Equations are conservative also in multiple dimensions.

Hyperbolicity

We cast the previous in quasilinear form

$$\frac{\partial \mathbf{Q}}{\partial t} + \mathbf{A}(\mathbf{Q}) \frac{\partial \mathbf{Q}}{\partial x} = \mathbf{S}(\mathbf{Q})$$

where $\mathbf{Q}$ is the vector of conserved variables and $\mathbf{A}(\mathbf{Q})$ is the quasilinear matrix, both given by

$$\mathbf{A}(\mathbf{Q}) = \left(\begin{array}{ccccc|c} 0 & \frac{1}{\tau} & 0 & 0 & 0 & \\ \alpha + 3c^2 - 1 & 0 & 0 & 0 & -\alpha & \\ 0 & 0 & 0 & -\gamma & 0 & \\ 0 & 0 & -\frac{1}{\beta} & 0 & 0 & \\ \hline & \mathbf{O}_{5,5} & & & & \mathbf{O}_{5,4} \end{array} \right),$$

$$\mathbf{Q} = (c, q_1, w, p_1, \varphi, q_2, q_3, p_2, p_3)^T,$$

Eigenvalues

System admits a full set of real eigenvalues given by

$$\chi_1 = -\frac{\sqrt{3c^2 + \alpha - 1}}{\sqrt{\tau}},$$

$$\chi_2 = -\frac{\sqrt{\gamma}}{\sqrt{\beta}},$$

$$\chi_{3-7} = 0,$$

$$\chi_8 = \frac{\sqrt{\gamma}}{\sqrt{\beta}},$$

$$\chi_9 = \frac{\sqrt{3c^2 + \alpha - 1}}{\sqrt{\tau}}.$$

and a corresponding set of linearly independent eigenfields.

Lyapunov Functional

Proposition

The proposed hyperbolic Cahn-Hilliard system admits the following Lyapunov functional

$$E = \int_{\mathcal{D}} e(c, \varphi, \mathbf{q}, \mathbf{p}, w) d\Omega,$$
$$e(c, \varphi, \mathbf{p}, w) = \frac{(c^2 - 1)^2}{4} + \frac{\gamma}{2} \|\mathbf{p}\|^2 + \frac{\alpha}{2} (c - \varphi)^2 + \frac{1}{2\beta} w^2 + \frac{1}{2\tau} \|\mathbf{q}\|^2$$

provided suitable boundary conditions are imposed. This is for instance the case for :

- (i) $\mathbf{p} \cdot \mathbf{n} = 0$, $\mathbf{q} \cdot \mathbf{n} = 0$, On $\partial\mathcal{D}$, where $\mathbf{n}$ is the normal vector exterior to the surface boundary $\partial\mathcal{D}$.*
- (ii) Periodic boundary conditions.*

Proof

We express the fluxes as a function of the conjugate variables

$$\frac{\partial c}{\partial t} + \operatorname{div} \left(\frac{\partial e}{\partial \mathbf{q}} \right) = 0$$

$$\frac{\partial \mathbf{q}}{\partial t} + \nabla \left(\frac{\partial e}{\partial c} \right) = - \frac{\partial e}{\partial \mathbf{q}}$$

$$\frac{\partial w}{\partial t} - \operatorname{div} \left(\frac{\partial e}{\partial \mathbf{p}} \right) = - \frac{\partial e}{\partial \varphi}$$

$$\frac{\partial \mathbf{p}}{\partial t} - \nabla \left(\frac{\partial e}{\partial w} \right) = 0$$

$$\frac{\partial \varphi}{\partial t} = \frac{\partial e}{\partial w}$$

Proof

We express the fluxes as a function of the conjugate variables

$$\frac{\partial e}{\partial c} \cdot \left\{ \frac{\partial c}{\partial t} + \operatorname{div} \left(\frac{\partial e}{\partial \mathbf{q}} \right) = 0 \right.$$

$$\frac{\partial e}{\partial \mathbf{q}} \cdot \left\{ \frac{\partial \mathbf{q}}{\partial t} + \nabla \left(\frac{\partial e}{\partial c} \right) = - \frac{\partial e}{\partial \mathbf{q}} \right.$$

$$\frac{\partial e}{\partial w} \cdot \left\{ \frac{\partial w}{\partial t} - \operatorname{div} \left(\frac{\partial e}{\partial \mathbf{p}} \right) = - \frac{\partial e}{\partial \varphi} \right.$$

$$\frac{\partial e}{\partial \mathbf{p}} \cdot \left\{ \frac{\partial \mathbf{p}}{\partial t} - \nabla \left(\frac{\partial e}{\partial w} \right) = 0 \right.$$

$$\frac{\partial e}{\partial \varphi} \cdot \left\{ \frac{\partial \varphi}{\partial t} = \frac{\partial e}{\partial w} \right.$$

Proof

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$$\frac{\partial e}{\partial c} \cdot \left\{ \frac{\partial c}{\partial t} + \operatorname{div} \left(\frac{\partial e}{\partial \mathbf{q}} \right) = 0 \right.$$

$$\frac{\partial e}{\partial \mathbf{q}} \cdot \left\{ \frac{\partial \mathbf{q}}{\partial t} + \nabla \left(\frac{\partial e}{\partial c} \right) = - \frac{\partial e}{\partial \mathbf{q}} \right.$$

$$\frac{\partial e}{\partial w} \cdot \left\{ \frac{\partial w}{\partial t} - \operatorname{div} \left(\frac{\partial e}{\partial \mathbf{p}} \right) = - \frac{\partial e}{\partial \varphi} \right.$$

$$\frac{\partial e}{\partial \mathbf{p}} \cdot \left\{ \frac{\partial \mathbf{p}}{\partial t} - \nabla \left(\frac{\partial e}{\partial w} \right) = 0 \right.$$

$$\frac{\partial e}{\partial \varphi} \cdot \left\{ \frac{\partial \varphi}{\partial t} = \frac{\partial e}{\partial w} \right.$$

$$\Rightarrow \frac{\partial e}{\partial t} + \operatorname{div} \left(\frac{\partial e}{\partial c} \frac{\partial e}{\partial \mathbf{q}} - \frac{\partial e}{\partial \mathbf{p}} \frac{\partial e}{\partial w} \right) = - \left\| \frac{\partial e}{\partial \mathbf{q}} \right\|^2 \leq 0,$$

Energy conservation: exchange form

$$e(c, \varphi, \mathbf{q}, \mathbf{p}, w) = e_I(c, \varphi, \mathbf{q}) + e_{II}(\mathbf{p}, w),$$

$$\begin{cases} e_I = \frac{(c^2 - 1)^2}{4} + \frac{\alpha}{2}(c - \varphi)^2 + \frac{1}{2\tau} \|\mathbf{q}\|^2, \\ e_{II} = \frac{\gamma}{2} \|\mathbf{p}\|^2 + \frac{1}{2\beta} w^2, \end{cases}$$

one can obtain the following evolution equation for each of the energy parts

$$\frac{\partial e_I}{\partial t} + \operatorname{div} \left(\frac{\partial e_I}{\partial c} \frac{\partial e_I}{\partial \mathbf{q}} \right) = - \frac{\partial e_I}{\partial \varphi} \frac{\partial e_{II}}{\partial w} - \left\| \frac{\partial e_I}{\partial \mathbf{q}} \right\|^2,$$

$$\frac{\partial e_{II}}{\partial t} - \operatorname{div} \left(\frac{\partial e_{II}}{\partial \mathbf{p}} \frac{\partial e_{II}}{\partial w} \right) = \frac{\partial e_I}{\partial \varphi} \frac{\partial e_{II}}{\partial w},$$

Numerical methods

In order to solve the model numerically and also compare it with reference solutions, we propose here:

- ① A numerical scheme for the original Cahn-Hilliard equation based on 4th order implicit conservative finite differences
- ② Explicit MUSCL-Hancock for the hyperbolic approximation.

Implicit conservative finite differences for CH

We propose here a semi-implicit conservative in order to solve numerically the original Cahn-Hilliard equations. We rewrite the latter as follows

$$\frac{\partial c}{\partial t} - \operatorname{div}(\mathbf{F}) + \gamma \Delta^2 c = 0$$

where $\mathbf{F}$ is the flux given by

$$\mathbf{F} = \chi(c) \nabla c, \quad \chi(c) = 3c^2 - 1$$

The scheme writes

$$c_{i,j}^{n+1} = c_{i,j}^n + \frac{\Delta t}{\Delta x} \left(\mathcal{F}_{i+\frac{1}{2},j}^{n+1} - \mathcal{F}_{i-\frac{1}{2},j}^{n+1} \right) + \frac{\Delta t}{\Delta y} \left(\mathcal{G}_{i,j+\frac{1}{2}}^{n+1} - \mathcal{G}_{i,j-\frac{1}{2}}^{n+1} \right) - \gamma \Delta \Delta_h c_{i,j}^{n+1}.$$

Computation of the intercell fluxes

The intercell fluxes $\mathcal{F}_{i+\frac{1}{2},j}^{n+1}$ and $\mathcal{G}_{i,j+\frac{1}{2}}^{n+1}$, in the x and y directions respectively, are computed using finite-differences as follows

$$\mathcal{F}_{i+\frac{1}{2},j}^{n+1} = \chi_{i+\frac{1}{2},j}^{n+1,r} (\nabla_x c)_{i+\frac{1}{2},j}^{n+1},$$

$$\begin{cases} \chi_{i+\frac{1}{2},j}^{n+1,r} \simeq \frac{1}{12} \left(7 \chi_{i,j}^{n+1,r} - \chi_{i-1,j}^{n+1,r} + 7 \chi_{i+1,j}^{n+1,r} - \chi_{i+2,j}^{n+1,r} \right) \\ (\nabla_x c)_{i+\frac{1}{2},j}^{n+1} \simeq -\frac{1}{12 \Delta x} \left(15 c_{i-1,j}^{n+1} - 15 c_{i,j}^{n+1} + c_{i+1,j}^{n+1} - c_{i+2,j}^{n+1} \right) \end{cases}$$

(similarly for $\mathcal{G}_{i,j+\frac{1}{2}}^{n+1}$)

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(similarly for $\mathcal{G}_{i,j+\frac{1}{2}}^{n+1}$)

These are 4th order approximations.

Discretization of the bi-laplacian operator

$\Delta\Delta_h c_{i,j}^{n+1}$ is a discretization of the bi-Laplacian operator in the cell-centers as follows

$$\begin{aligned}\Delta\Delta_h c_{i,j}^{n+1} = & -\frac{\Delta t}{\Delta x^4} \left(c_{i-2,j}^{n+1} - 4c_{i-1,j}^{n+1} + 6c_{i,j}^{n+1} - 4c_{i+1,j}^{n+1} + c_{i+2,j}^{n+1} \right) \\ & -\frac{\Delta t}{\Delta y^4} \left(c_{i,j-2}^{n+1} - 4c_{i,j-1}^{n+1} + 6c_{i,j}^{n+1} - 4c_{i,j+1}^{n+1} + c_{i,j+2}^{n+1} \right) \\ & -\frac{2\Delta t}{\Delta x^2 \Delta y^2} \left(c_{i-1,j-1}^{n+1} - 2c_{i,j-1}^{n+1} + c_{i+1,j-1}^{n+1} - 2c_{i-1,j}^{n+1} \right. \\ & \quad \left. + 4c_{i,j}^{n+1} - 2c_{i+1,j}^{n+1} + c_{i-1,j+1}^{n+1} - 2c_{i,j+1}^{n+1} + c_{i+1,j+1}^{n+1} \right)\end{aligned}$$

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System over all the domain is then solved using GMRES (Matrix is not symmetric positive-definite ...)

Numerical method for hyperbolic approximation

- Explicit second-order MUSCL-Hancock scheme
- We used FORCE / Rusanov approximate Riemann solvers (One could also implement a Roe solver)

Convergence in α : ODE for original system

CH equation in 1D:

$$\frac{\partial c}{\partial t} + \frac{\partial J}{\partial x} = 0, \quad J = \gamma \frac{\partial^3 c}{\partial x^3} - (3c^2 - 1) \frac{\partial c}{\partial x}$$

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Stationary states:

$$\begin{aligned} \frac{\partial J}{\partial x} &= 0, \quad J = \gamma \frac{\partial^3 c}{\partial x^3} - (3c^2 - 1) \frac{\partial c}{\partial x}, \\ J(x_0) &= J^0, \quad c(x_0) = c^0, \quad \left. \frac{\partial c}{\partial x} \right|_{x=x_0} = c_I^0, \quad \left. \frac{\partial^2 c}{\partial x^2} \right|_{x=x_0} = c_{II}^0, \end{aligned}$$

which can be written in a first-order system as

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which can be written in a first-order system as

$$\begin{aligned} c' &= c_I, \quad c_I' = c_{II}, \quad c_{II}' = \frac{1}{\gamma} (J + (3c^2 - 1)c_1), \quad J' = 0, \\ c(x_0) &= c^0, \quad c_I(x_0) = c_I^0, \quad c_{II}(x_0) = c_{II}^0, \quad J(x_0) = J^0, \end{aligned}$$

Hyperbolic counterpart

$$\begin{aligned} \varphi' &= p, \quad p' = \frac{\alpha}{\gamma} (\varphi - c), \quad c' = \frac{\alpha p - q/\tau}{3c^2 - 1 + \alpha}, \quad q' = 0, \\ \varphi(x_0) &= c^0 + \frac{\gamma}{\alpha} c_{II}^0, \quad p(x_0) = c_I^0, \quad c(x_0) = c^0, \quad q(x_0) = -\tau J^0, \end{aligned}$$

Reminder about original system ODE

$$\begin{aligned} c' &= c_I, \quad c_I' = c_{II}, \quad c_{II}' = \frac{1}{\gamma} (J + (3c^2 - 1)c_1), \quad J' = 0, \\ c(x_0) &= c^0, \quad c_I(x_0) = c_I^0, \quad c_{II}(x_0) = c_{II}^0, \quad J(x_0) = J^0, \end{aligned}$$

Convergence in α

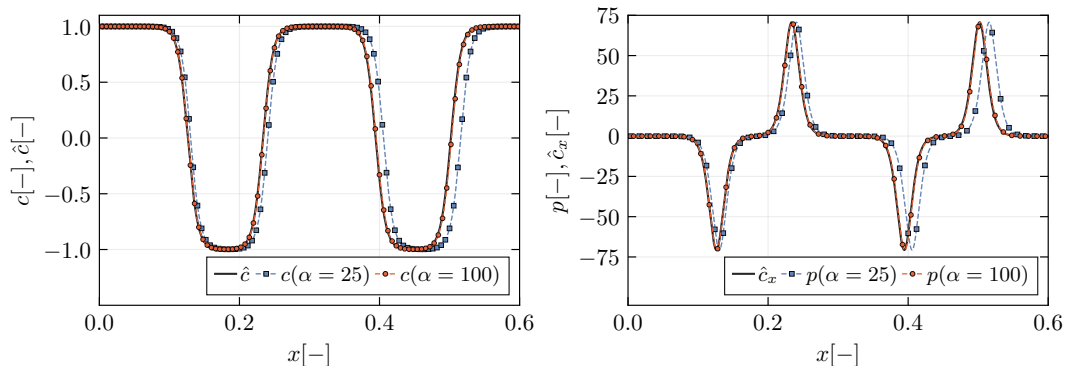


Figure 1: Comparison of a stationary solution of the hyperbolic Cahn-Hilliard model (discontinuous lines) with the original counterpart (solid line) for different values of the penalty parameter α .

Convergence table in α

α	$\ c - \hat{c}\ _{L_2}$	$\ \mathbf{p} - \nabla \hat{c}\ _{L_2}$	$\ c - \varphi\ _{L_2}$	$\mathcal{O}(c - \hat{c})$	$\mathcal{O}(\mathbf{p} - \nabla \hat{c})$	$\mathcal{O}(c - \varphi)$
25	2.64×10^{-1}	5.66×10^{-1}	7.01×10^{-3}	—	—	—
50	1.35×10^{-1}	3.02×10^{-1}	3.51×10^{-3}	0.96	0.90	0.99
100	6.82×10^{-2}	1.54×10^{-1}	1.75×10^{-3}	0.99	0.97	0.99
400	1.70×10^{-2}	3.86×10^{-2}	4.39×10^{-4}	1.00	0.99	0.99
1600	3.80×10^{-3}	8.64×10^{-3}	1.10×10^{-4}	1.08	1.08	1.00

Table 1: Convergence table for the L_2 errors when comparing the numerical Cauchy problem solutions for the hyperbolic model with the the original Cahn-Hilliard equation.

Exact solution for the original equation

One can find a family of exact one-dimensional stationary periodic solutions to the Cahn-Hilliard system expressed as

$$c_{\epsilon}(x) = \sqrt{1-\epsilon} \operatorname{sn} \left(\sqrt{\frac{\epsilon+1}{2\gamma}}(x-x_0), \sqrt{\frac{1-\epsilon}{1+\epsilon}} \right)$$

Here, $\operatorname{sn}(x, s)$ is the Jacobi elliptic sine function, and s is the elliptic modulus.
 $\epsilon \in [0, 1]$.

It is worthy of note that in the limit $\epsilon \rightarrow 0$ corresponding to $s \rightarrow 1$, one recovers the well-known solution

$$c(x) = \tanh \left(\frac{x-x_0}{\sqrt{2\gamma}} \right)$$

as a particular case.

Exact elliptic function solution

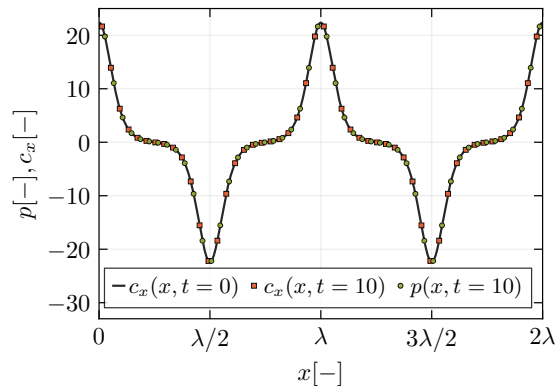
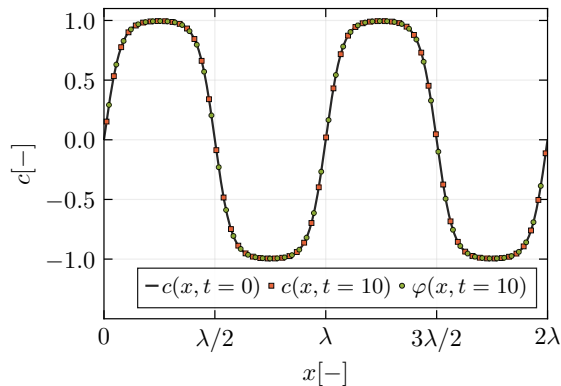


Figure 2: $\gamma = 0.001$. Computational domain is $[0, 2\lambda]$, discretized over $N = 2000$ cells. $\beta = 10^{-6}$, $\alpha = 500$ and $\tau = 8 \cdot 10^{-4}$. CFL = 0.95 and final simulation time is $t = 10$.

Spinodal decomposition ($\gamma = 10^{-3}, \beta = 10^{-7}, \alpha = 500, \tau = 10^{-5}$)

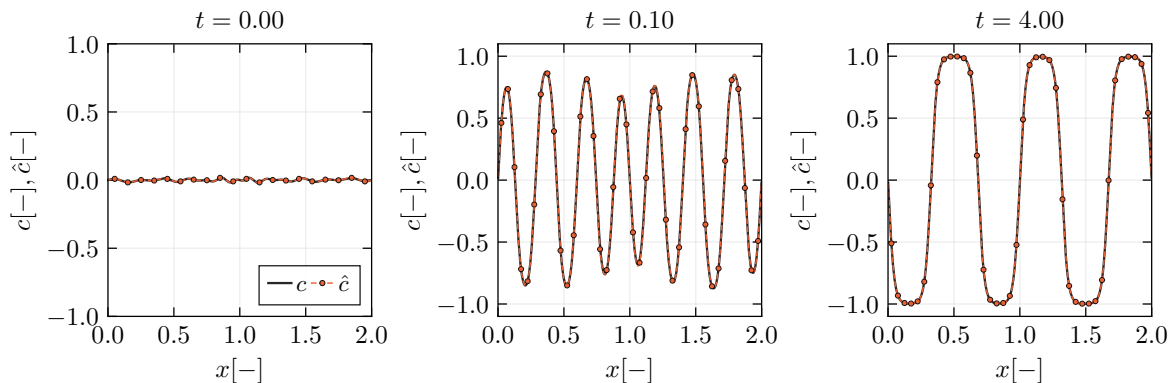


Figure 3: Comparison of the numerical results between the original model (orange) and its hyperbolic counterpart (black). $N = 2000$ computational cells.

Spinodal decomposition - Lyapunov function evolution ($\simeq 6b$ iterations)

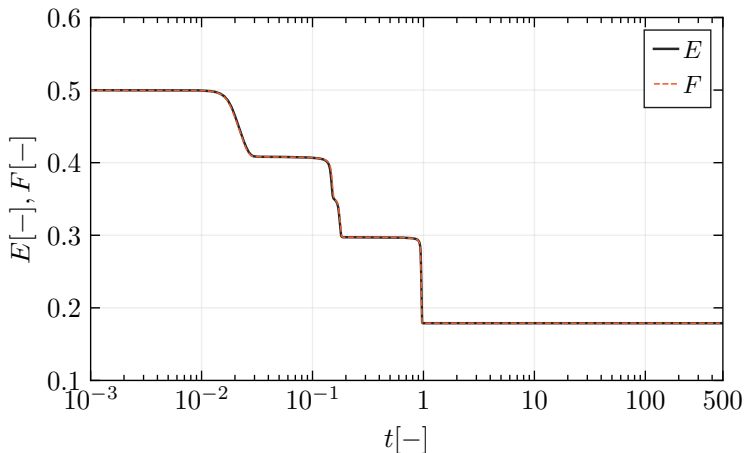


Figure 4: Lyapunov function evolution over time (F :original, E : hyperbolic.)

Lyapunov function evolution ($\simeq 30b$ iterations)

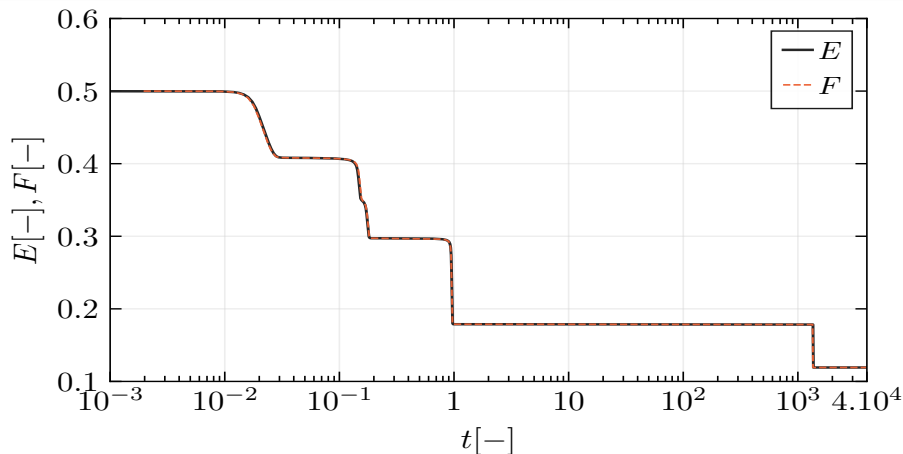


Figure 5: Lyapunov function evolution over time (F :original, E : hyperbolic.)

Ostwald Ripening in 1D ($\gamma = 10^{-3}, \beta = 10^{-7}, \alpha = 500, \tau = 10^{-4}$)

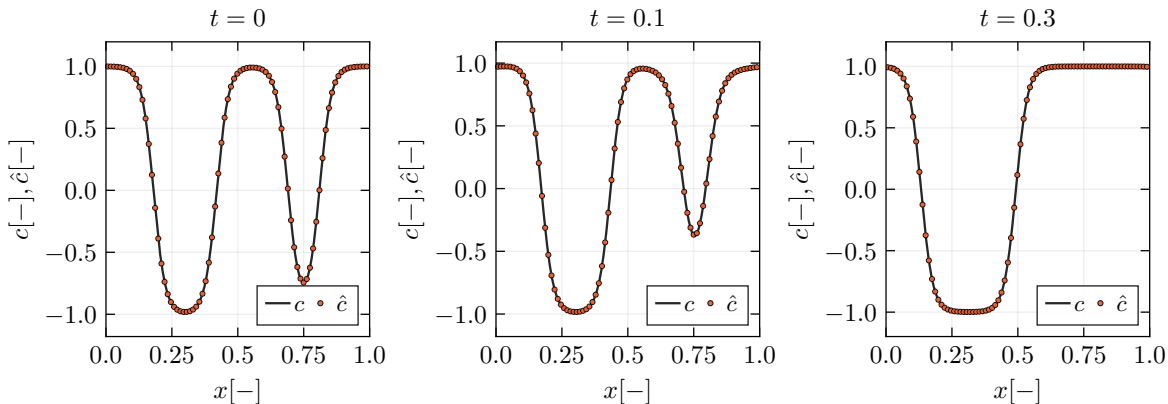


Figure 6: Comparison of the numerical solutions for hyperbolic Cahn-Hilliard model (black line) and the original model (red dots) for the Ostwald Ripening test case at times $t = \{0, 0.1, 0.3\}$.

Stationary Bubble

we numerically solve the time-dependent Cahn-Hilliard equation in radial coordinates

$$\frac{\partial c}{\partial t} - \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial}{\partial r} \left(c^3 - c - \frac{\gamma}{r} \frac{\partial}{\partial r} \left(r \frac{\partial c}{\partial r} \right) \right) \right) = 0, \quad r = \sqrt{x^2 + y^2},$$

and we consider the following initial guess

$$c_0(r) = -\tanh \left(\frac{r - 0.5}{\sqrt{2\gamma}} \right),$$

We use the same scheme presented earlier for the original Cahn-Hilliard equations.

Stationary Bubble

we numerically solve the time-dependent Cahn-Hilliard equation in radial coordinates

$$\frac{\partial c}{\partial t} - \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial}{\partial r} \left(c^3 - c - \frac{\gamma}{r} \frac{\partial}{\partial r} \left(r \frac{\partial c}{\partial r} \right) \right) \right) = 0, \quad r = \sqrt{x^2 + y^2},$$

and we consider the following initial guess

$$c_0(r) = -\tanh \left(\frac{r - 0.5}{\sqrt{2\gamma}} \right),$$

We use the same scheme presented earlier for the original Cahn-Hilliard equations. (Pseudo-transient continuation methods).

Stationary bubble($\gamma = 10^{-3}, \beta = 10^{-7}, \alpha = 1000, \tau = 10^{-5}$)

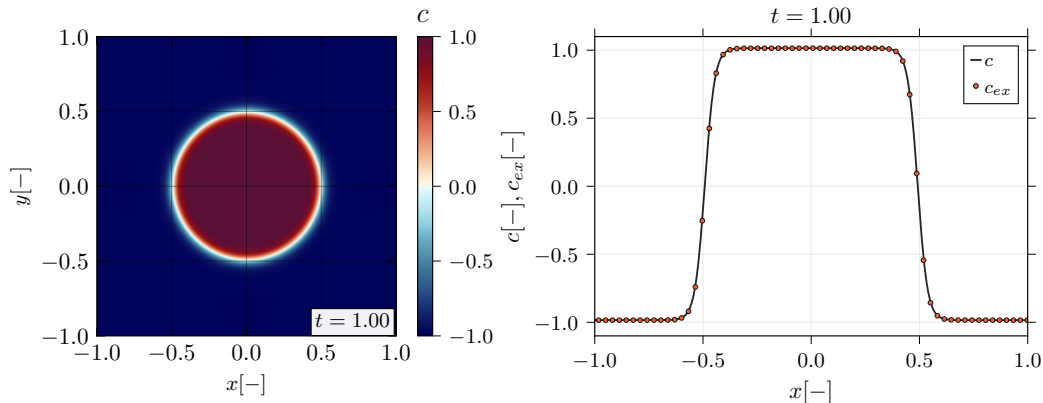
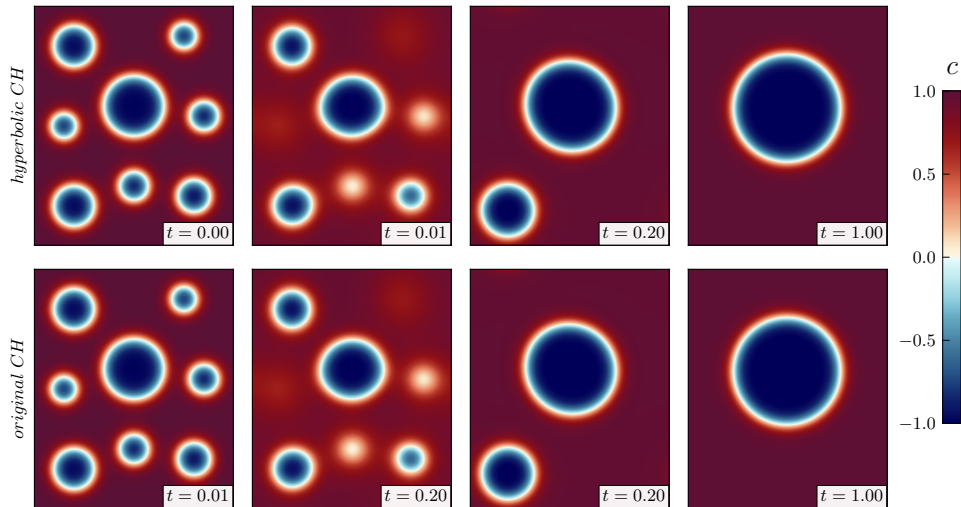


Figure 7: Left: 2D color plot of the radial stationary solution. Right: radial cut of the numerical solution along the line $y = 0$ at $t = 1$ (black line), compared with the exact solution c_{ex} , provided as initial data (red dots). Domain is 500×500

Ostwald Ripening in 2D($\gamma = 10^{-3}, \beta = 10^{-7}, \alpha = 500, \tau = 10^{-5}$)



Ostwald Ripening in 2D : horizontal Cuts

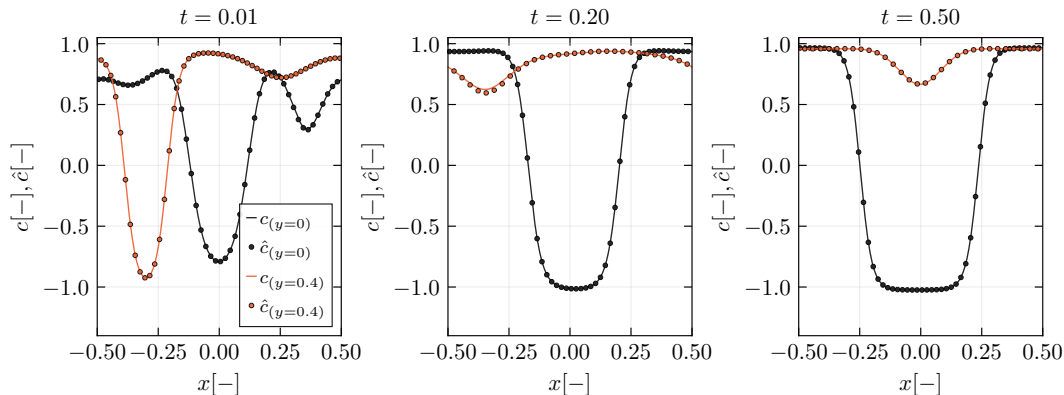


Figure 8: Horizontal cuts over the lines $y = 0$ (red) and $y = 0.4$ (black). Domain is 600×720

Conclusion and Perspective

- I presented a new formulation for an approximate hyperbolic Cahn-Hilliard system.
- An original scheme was conceived to solve the original equation using conservative finite differences.
- Comparison of results showed excellent agreement between the results in one and two dimensions.

Perspectives

- Extension to Navier-Stokes Cahn-Hilliard systems.
- Sharp interface limit, admissible shocks, contacts ?
- Proof of convergence in the limit of relaxation.
- More clear BC/IC correspondence.
- Semi-implicit discretization, asymptotic preserving schemes, time-step optimization, etc...

Thank you for your attention !

[1] Dhaouadi, Firas, Michael Dumbser, and Sergey Gavriluk. "A first-order hyperbolic reformulation of the Cahn–Hilliard equation." *Proceedings A*. Vol. 481. No. 2312. The Royal Society, 2025.

[2] Dhaouadi, Firas; Dumbser, Michael; Gavriluk, Sergey (2025). Supplementary material from "A first-order hyperbolic reformulation of the Cahn-Hilliard equation". The Royal Society. Collection.

Acknowledgement: *This project was supported and funded by NextGeneration EU, Azione 247 MUR Young Researchers – MSCA/SoE.*

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If you are seeing this, you must have asked about how to choose the Boundary conditions / Initial conditions...